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Where can we hear silence? Exploring spatial patterns of perceptually-defined silent locations

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ABSTRACT

The identification of quiet areas is currently a national obligation of EU countries. Factors that promote tranquillity go beyond low-noise exposure levels and may depend on context and type of noise sources, composing each space. This study applied the subjective, listener-centred ISO 12913-2 soundscape methodology to survey the auditory experience of the Flemish population in Belgium. Participants were asked to what extent they perceived silence. Out of 4665, N=361 respondents reported hearing it “a lot” or “predominantly”. We set out to characterize these silent environments using geospatial features to investigate potential spatial patterns that may drive this experience. As expected, the area perceived as silent are also more calm and pleasant, featuring natural sounds. Most of those 361 answers came from locations with lower population density and at least 2 km away from a highway. Moreover, the more built-up the surrounding environment was, the less the individuals perceived silence. Lastly, we investigated the answers from the 18 soundwalks conducted within the officially designated quiet areas in the open country. Although few rated them as silent, a higher percentage found the acoustic environment be very calm and pleasant, suggesting that they are still considered places of potential relaxation.

Keywords: *soundscape, soundwalk, quiet area, GIS, surveys.*

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1. INTRODUCTION

The growing recognition of environmental noise as a major health hazard has led to the widespread adoption of legislative measures in Europe, most notably the Environmental Noise Directive (END) (Directive 2002/49/EC). While noise pollution is often approached with a focus on reducing the associated burdens, the benefits of living in/visiting quiet areas are relatively understudied [1]. In this sense, the END highlights the need to preserve environmental noise quality “where it is good” [2].

Quiet areas in the open country are defined as those undisturbed by noise from traffic, industry or recreational activities, but they also exist inside agglomerations. While Member States use actual noise levels to define quiet areas, either based on noise modelling or actual acoustic measurements in situ, the evaluation of user/visitor experiences is also considered. In this sense, the soundscape approach (ISO 12913 series) is an instrument to quantify the subjective perception of acoustic environments and has been combined with the methodologies to define quiet areas.

This work explores the results from an extensive soundscape survey launched by the citizen science project *De Oorzaak* in Flanders (Belgium) (see [3] for more details). Citizens were invited to go for a soundwalk on their own and answer a questionnaire based on ISO 12913-2. The survey was publicized through the press, resulting in large public engagement and geographical coverage spanning the entirety of Flanders (13,625 km²). Besides declaring the extent to which they heard common noise sources (traffic, natural sounds, etc.), participants were asked to what extent they experienced silence.

This study aims to answer the following questions: 1) How does perceiving silence during a soundwalk relate to other





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perceptual attributes? 2) How are geospatial characteristics of locations linked to the perception of silence? and 3) Do perceptually defined silent locations align with officially designated quiet areas?

2. MATERIALS AND METHODS

2.1 Data collection and survey content

The data collection started on 20/10/2023 via convenience sampling using a Qualtrics survey. Most responses were received during the initial weeks following the survey release via the press (newspaper De Morgen) and the University of Antwerp's communication channels.

The participants were asked to go outdoors on a soundwalk for 2-5 minutes, alternate between walking and standing still while actively listening and observing the environment, and then complete the questionnaire. Proceeding to the questions was only possible upon indicating to be at least 18 years old and consent to the data processing procedures. The Ethics Committee for the Social Sciences and Humanities of the University of Antwerp granted ethical approval for the study on September 18th, 2023 (Ref No: SHW_2023_215_1). The questionnaire was based mainly on ISO/TS 12913-2, Method A. The relevant questions for this paper comprise:

- Noise source identification: On a 5-point scale from “not at all” to “dominates completely”, participants indicated to what extent the following noise sources were heard: traffic, sounds from human beings, music, nature sounds, alarms/priority vehicles, industry noise, construction and/or maintenance works, silence.
- Perceived affective quality: The extent to which the acoustic environment was characterized according to eight attributes (eventful–uneventful, vibrant–monotonous, calm–chaotic, and annoying–pleasant) was assessed using a 5-point scale ranging from “totally disagree” to “totally agree.”
- Approximate address where the soundwalk was conducted.

2.2 Geospatial features and data analysis

GPS locations were retrieved through the Google Maps API from the provided addresses. Open-source databases were resourced to characterize their surroundings, described as follows.

The population density was obtained at two levels: 1) at the municipality level, using data available from the Flemish government, and 2) at the statistical sector level, a

subdivision of a municipality, which is the most detailed territorial level provided by the Belgian Statistical Office.

To quantify green space, we used the Groenkaart Vlaanderen 2021 [4], a 1-m resolution raster dataset derived from orthorectified aerial imagery, which classifies each pixel into four classes: not green, agriculture, low green (vegetation under 3 m, including non-agricultural grasslands, meadows, shrubs, and small trees), or high green (vegetation at least 3-m tall). The fraction of each class was calculated as the covered area to the total buffer area of a 500-m radius around the provided locations, assuming values between 0 and 1.

Using QGIS and Open Street Map data, we obtained the length of main roads (categories: motorway, primary, secondary, and tertiary) within the 500-m buffer area and calculated the total weighted road length by summing the lengths multiplied by weighing factors depending on the road category (e.g., 4 for motorway, 3 for primary, etc.). We also measured the shortest distance from the participants' locations to each road class.

Lastly, the Quietness Suitability Index (QSI) at each location was retrieved from the 100-m resolution European Environmental Agency's (EEA) dataset [5] (last update: July 2024). Additionally, we compared them to those officially defined as quiet areas by the Flemish government [6] (last update: September 2024).

The initial sample was split into two groups: those who reported hearing “not at all” (score 1 out of 5) and those who heard “a lot” or “predominantly” silence (score 4 or 5 out of 5). We evaluated the difference between these groups over some variables: the soundscape attributes, the sound sources identification, the QSI values, population density, amount of greenery, and distance to roads and road total weighted length. Mean comparison tests were used to check for differences at a 1% significance level.

3. RESULTS

3.1 Sample

In total, 4665 complete answers were recorded. Figure 1 illustrates their geographical distribution, colour-coded based on how much the respondents heard silence. The map also delineates the boundaries of Belgian municipalities, coloured according to the five population density classes defined in the Written Environmental Survey by the Flemish government (SLO-4 [7]): Category 1: <300, Category 2: 301-600, Category 3: 601-1000, Category 4: 1001-3000, Category 5: >3000 inhabitants/km². These are



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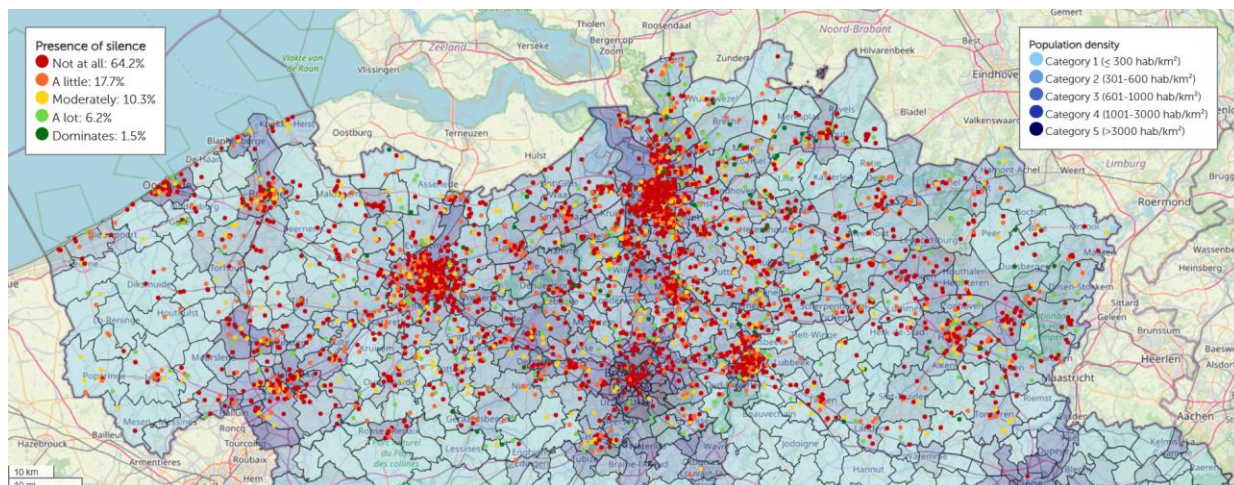


Figure 1 Geographical spread of the survey responses (N= 4665).

further narrowed down into rural (Category 1-2) and urban (Category 3-5).

Participants from urban areas (Categories 3-5) accounted for 67.3%, while rural areas comprised a significant 1/3 of the sample (32.6%). Out of the 4665 answers, 2994 reported not hearing silence at all (red dots), while 361 heard at least a lot of silence (green dots). Therefore, a sample of N=3355 will be used in the subsequent analyses.

3.2 The soundscape of perceived silent areas

Figure 2 presents the average scores of soundscape dimensions for perceiving a lot of silence or not at all. Mann-Whitney U tests revealed significant differences between the two groups across all attributes.

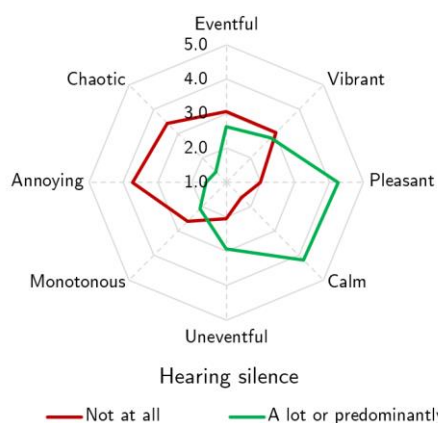


Figure 2. Radar plots for average soundscape attributes per “hearing silence” group.

Notably, silent acoustic environments are found in the pleasant-calm-uneventful zone of the circumplex model, whereas those judged as not silent tend to align with the eventful-chaotic-annoying region. Interestingly, both groups exhibit similar scores for vibrant, which is a dimension thought to have a positive connotation. Furthermore, the average monotonous score is higher for non-silent environments, indicating that the absence of prominent (and likely unpleasant) sound sources does not necessarily lead to a perception of boredom.

3.3 Silence and sound sources

Figure 3 indicates the average degree to which sound sources were heard, including silence, for both categories of perceived silence.

Mann-Whitney U tests indicate statistically significant differences across all sources. Noise from industry and alarms/priority vehicles, along with music, score low for both categories, likely because these are, in any case, restricted to particular events and locations.

The most pronounced differences involved natural sounds and traffic. Traffic seems to be the main disruptor of silence (average 4.04 out of 5 for the “not at all” silent group); however, even in environments deemed silent, traffic was still perceptible to some extent (2.35 out of 5), suggesting that background-like traffic noise does not fully compromise perceived silence. Silent environments score considerably higher in natural sounds (3.45 against 2.50 for not silent), reinforcing the notion that quietness is not necessarily linked to low noise levels, but to soothing, pleasant acoustic environments.



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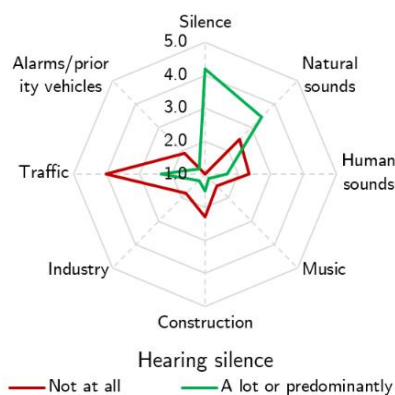


Figure 3. Radar plots for average noise source intensity per “hearing silence” group.

In the survey by [8], participants also rated the quiet rural soundscape as dominated by nature sounds but rather in harmony with the environment, which underlines the quiet character of it.

3.4 The effect of population density

Figure 4 shows the percentage of responses from the hearing “a lot/predominantly”-silence answers, broken down by the four population density categories (at the municipality level). Note that the sample used in this analysis considers only answers from the categories “not at all” and “a lot”/ “predominantly” hearing silence (N=3355). The annotations by the data points indicate the actual number of answers. Among all answers from areas in Category 1, 25.5% reported hearing a lot of silence. This value consistently reduces for the more densely populated categories, reaching 3.8% in Category 4. The number of answers in Category 5 is almost negligible, as very few municipalities belong to this category in Flanders.

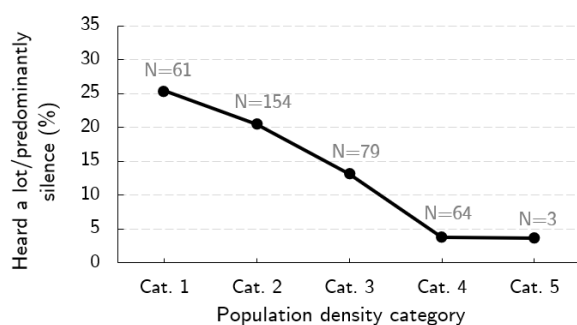


Figure 4. Percentage of respondents perceiving a lot/predominantly silence by population density category.

3.5 The impact of vegetation on perceived silence

Figure 5 introduces the percentage of responses reporting a lot/predominantly silence, segmented by 10 levels of vegetation fraction within the buffer area. For example, a value of 0.5 indicates that 50% of the buffer area is covered by a given land use. The annotations indicate the actual number of answers. While three vegetation types are analyzed, the “low green” category is omitted as insufficient instances at higher ratios (above 0.5) limited the relevance of this analysis.

The percentage of the silence perceiving group increases as the fractions of high green or agricultural land increase. In areas with minimal coverage (0–0.1), fewer than 10% of respondents noted mostly silence. In contrast, in zones where high green or agriculture occupies 60–70% of the area, these figures increase to 22% and 32%, respectively. Notably, agricultural land cover consistently results in higher levels of perceived silence than high green.

The non-green areas reveal an even clearer trend, but naturally in the opposite direction. In regions with almost no non-green land (0–0.1), 44% reported a lot of silence. As the proportion of non-green land increases, this percentage declines sharply, dropping below 10% at 0.4–0.5 and higher.

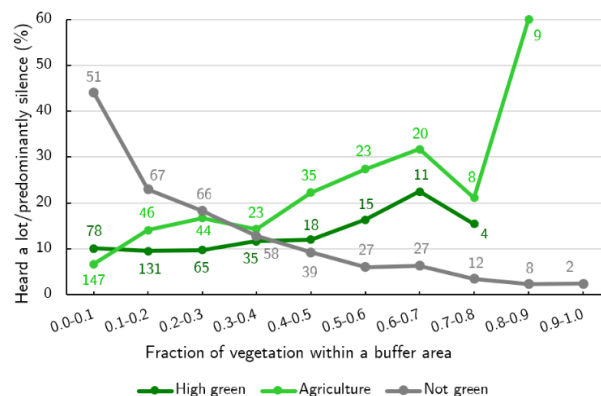


Figure 5. Percentage of respondents perceiving a lot/predominantly silence by vegetation cover fraction in the buffer area.

3.6 Impact of traffic on perceived silence

Figure 6 presents boxplots illustrating the distribution of distances from each participant to the closest motorway, primary, secondary and tertiary road, split by perceiving or not silence. Clearly, soundwalks taken in perceived silent acoustic environments were farther from roads. For



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example, about 75% of these answers are at least 2 km away from a highway. T-tests confirmed statistically significant differences between the two groups for all road categories. However, these differences diminish as we move lower in the road hierarchy – to be expected, as lower speeds and traffic volumes result in less road traffic noise emission.

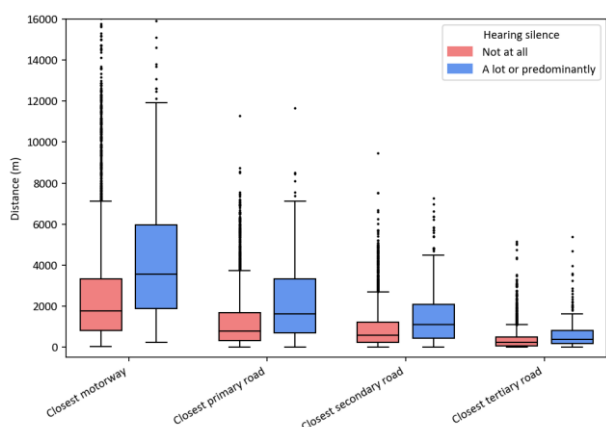


Figure 6. Boxplots of distances to the nearest road, per road category.

Figure 7 displays the percentage of “hearing silence” responses according to the weighted total road length within the buffer area around the participants, grouped in 500-m increments (except for the category None and >5000m). The annotations indicate the actual number of answers.

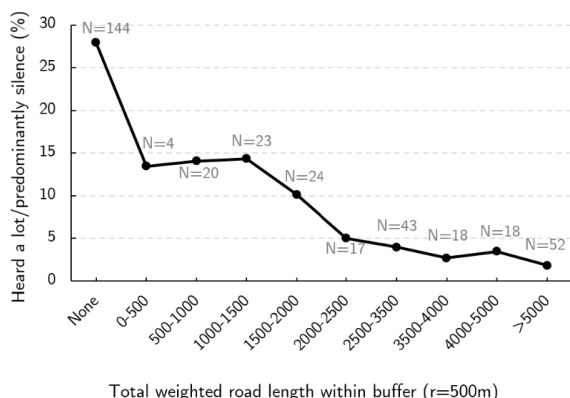


Figure 7. Percentage of respondents perceiving a lot/predominantly silence by weighted total road length within buffer area.

Almost 40% (N=144) of respondents who reported a lot of silence had no roads within their immediate 500-m surroundings. As road density increases, the proportion of this group consistently decreases. Beyond 2000 m of total weighted road length, fewer than 5% reported perceiving silence. These results underscore the strong influence of traffic on the perception of silence.

3.7 Relations among population density, traffic, and vegetation

From the geospatial features analyzed in Sections 3.4 to 3.6, population density, lack of greenery and distance to roads, especially highways, showed the most apparent impact on perceived silence. To further observe these patterns, Figure 8 plots these three variables together. Note that the population density here is taken at the statistical sector level instead of the municipality level for more granular values. The data points only occupy a part of the 3D space since all three variables act as proxies of urbanization degree. Higher population density leads to higher non-green fractions. Similarly, high non-green fractions also coincide with locations closer to a highway, and so on. Additionally, Belgium, particularly the Flemish region, has a dense road network, resulting in few points at long distances to a motorway.

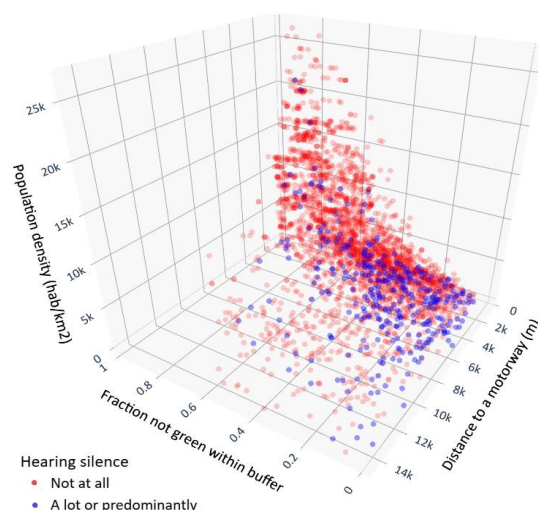


Figure 8. Three-dimensional scatter plot of population density, fraction not green, and distance to the nearest motorway ([interactive plot link](#)).

Perhaps the most noticeable difference between the red (not-at-all silent group) and blue (a lot/predominantly silent



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group) data points is found along the population density axis, with the blue dots mainly occupying the less densely populated areas. It is also clear that a large portion of the blue dots are located in less built-up areas and tend to be farther from highways than the red dots.

Given the large sample size of the non-silence group, red dots are spread all along the fraction not-green axis. The red dots in less built-up areas (lower fraction not green) also tend to be close to a motorway, which likely reduced the positive impact that being surrounded by nature could have on the perception of silence. Conversely, many blue dots are also located in similar settings (near motorways and largely unbuilt). This observation indicates a complex dynamic: while for some, nature may enhance the perception of silence even if close to a highway, this effect is not universally experienced by all participants.

3.8 Perceived silence vs quiet areas in Flanders

To compare the perceived silent locations from the soundwalks and the officially defined quiet areas in Flanders, we drew on two databases. First, in Figure 8, the layer covering the entire region in red-to-blue hues displays the QSI values obtained from the EEA map [5]. A value of 1 denotes a suitable quiet area, and 0 is completely unsuitable. We extracted the QSI score from this layer at each participant's location. Second, the polygons in black represent the silent areas defined by the Flemish government [6], which uses two definitions: the solid-filled contours are the "Quality Label Quiet Areas", and the X-patterned indicate the acoustically-surveyed potential quiet areas. Although both methodologies (Flanders and EEA) use similar criteria to define quietness — considering not only noise levels but distances to noise sources, natural

landscape, etc. — these do not overlap completely. Note: these datasets only account for non-urban zones.

As highlighted in the EEA report [9], Belgium has one of the smallest proportions of quiet areas among all Member States, mainly due to its high population density. This shortage is even more pronounced in Flanders, as seen in the predominantly red patterns of Figure 8. The average QSI for participants who heard "not at all" silence was 0.0018, whereas those who reported hearing a lot/predominantly silence were in locations with an average QSI about ten times higher, though still only 0.017. A t-test confirmed these differences were statistically significant.

From the complete dataset (N=4665), only 18 soundwalks were conducted in the silent areas defined by the Flemish government. From these, only 27.7% reported hearing a lot/predominantly silence, but 50% found the place to be very calm (score 4 or 5 out of 5), and 66.7% found it very pleasant (4 or 5 out of 5).

4. CONCLUSION

This study explored the results from a region-wide soundscape survey conducted through a citizen science project (*De Oorzaak*). The large sample size and geographical spread allowed us to focus on a specific aspect: the perception of silence. Our analyses compared the 361 answers who reported hearing a lot/predominantly silence to those 2994 who did not perceive silence at all. The answers to the research questions are detailed below.

RQ1: How does perceiving silence during a soundwalk relate to other perceptual attributes?

Perceptually silent environments enable a calm, pleasant,

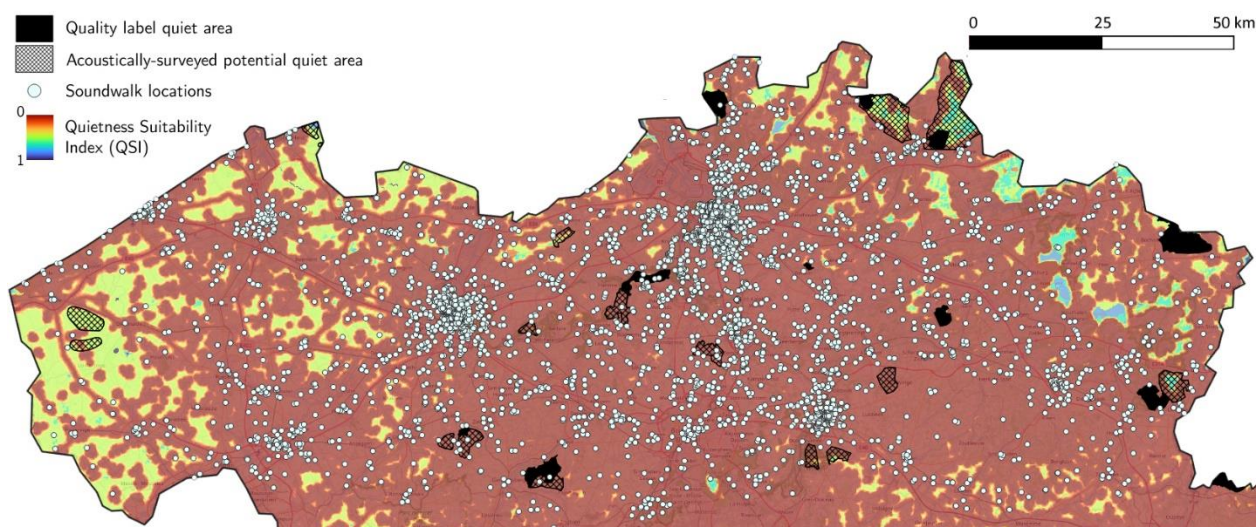


Figure 9. QSI values, officially defined quiet areas in Flanders, and soundwalk locations.



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and uneventful soundscape, while those rated as “not at all” silent were mostly judged chaotic, annoying, and eventful. Both groups scored similarly in vibrant dimension, yet non-silent environments were significantly more monotonous. This may suggest that prominent noise sources do not necessarily foster engagement or variety in the auditory experience; instead, if these sources are considered unpleasant or intrusive, they may actually take attention that would otherwise be paid to potentially stimulating aspects of the acoustic environment.

Concerning the identification of noise sources, traffic noise is strongly present in non-silent environments, with an average rating of 4.04 out of 5. Nonetheless, traffic noise is often still noticeable in perceived silent environments but at a lower intensity (average 2.35), which suggests that background-like traffic noise may not completely disrupt the positive perception of silence. Additionally, those who experienced a lot of silence also reported considerably more natural sounds (3.45 compared to 2.50 in non-silent areas), corroborating the idea that quiet areas are valued for their positive appraisals and restorative qualities, not just for lower noise levels.

RQ2: How are geospatial characteristics of locations linked to the perception of silence?

We investigated three geospatial features: population density, presence of vegetation, and distance to a road and road density.

Although about 2/3 of the soundwalks were conducted in urban environments, around 60% of those who perceived a lot of silence were located in rural zones (<600 inhab/km²). Also, a decreasing share of this group is observed with increasing population density. Distance to roads and road density around participants also strongly reflected this perception. Participants from the silence group were statistically significantly farther from all categories of roads than those who did not perceive silence. These differences, however, decreased for smaller road categories, to be expected given the lower traffic volumes and speeds. Concerning road density, almost 40% of respondents who heard a lot of silence had no roads whatsoever within their immediate 500-m surroundings, and as road density increased, the proportion of this group consistently decreased compared to the percentage who didn't hear silence.

These results demonstrate that locations perceived as silent in our soundwalks conform to the formal definition of quiet areas: undisturbed by human-made noises such as from traffic, industry, or recreational activities [9].

High vegetation and agricultural land within the 500 m surroundings of the soundwalk locations markedly impacted the perception of silence. This effect was the largest for agricultural land, likely linked to a lower population density. Conversely, increased non-green cover, indicative of built-up areas, led to a significant exponential decrease in the share of participants perceiving silence.

RQ3: Do perceptually defined silent locations align with officially designated quiet areas?

Belgium, and especially Flanders, has a notably low proportion of quiet areas to its territory. From our sample, only 18 soundwalks were conducted within the official quiet zones. These were not necessarily rated quiet – only 27.7% perceived a lot/predominantly silence. However, a higher percentage found these areas to be very calm and pleasant, suggesting that while they might not be directly perceived as quiet, they are still considered places of potential relaxation by the visitors. Additionally, the Quietness Suitability Index (QSI) for both perceived silent and non-silent areas are remarkably low – 0.017 and 0.0018 on average, respectively, on a 0-1 scale. Yet, the former is ten times higher than the latter, and this difference proved to be statistically significant.

5. ACKNOWLEDGMENTS

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